Feture Engineering

Step 1: Import Libraries

```
import pandas as pd
import numpy as np
```

Load the data

```
blinkit = pd.read_csv("blinkit_feedback.csv")
```

Step 2: Convert Date Column

```
blinkit["feedback_date"] = pd.to_datetime(blinkit["feedback_date"])
```

Check

```
blinkit.dtypes
```

Step 3: Create Date Features

Year

```
blinkit["Year"] = blinkit["feedback_date"].dt.year
```

Month

```
blinkit["Month"] = blinkit["feedback_date"].dt.month
```

Month Name

```
blinkit["Month_Name"] = blinkit["feedback_date"].dt.month_name()
```

Day

```
blinkit["Day"] = blinkit["feedback_date"].dt.day
```

Day of Week

```
blinkit["Day_Name"] = blinkit["feedback_date"].dt.day_name()
```

Week Number

```
blinkit["Week"] = blinkit["feedback_date"].dt.isocalendar().week
```

Quarter

```
blinkit["Quarter"] = blinkit["feedback_date"].dt.quarter
```

Weekend Feature

```
blinkit["Weekend"] = blinkit["feedback_date"].dt.dayofweek >= 5
```

Convert Boolean to Integer

```
blinkit["Weekend"] = blinkit["Weekend"].astype(int)
```

Step 4: Rating Category

Instead of using only ratings

```
def rating_category(x):
    if x >= 4:
        return "Good"
    elif x == 3:
        return "Average"
    else:
        return "Poor"

blinkit["Rating_Category"] = blinkit["rating"].apply(rating_category)
```

Step 5: Feedback Length

```
blinkit["Feedback_Length"] = blinkit["feedback_text"].str.len()
```

Step 6: Word Count

```
blinkit["Word_Count"] = blinkit["feedback_text"].str.split().str.len()
```

Step 7: Character Count

```
blinkit["Character_Count"] = blinkit["feedback_text"].str.len()
```

Step 8: Number of Sentences

```
blinkit["Sentence_Count"] = blinkit["feedback_text"].str.count(r"\.") + 1
```

Step 9: Capital Letters Count

```
blinkit["Capital_Letters"] = blinkit["feedback_text"].str.count(r"[A-Z]")
```

Step 10: Exclamation Marks

```
blinkit["Exclamation"] = blinkit["feedback_text"].str.count("!")
```

Step 11: Question Marks

```
blinkit["Question"] = blinkit["feedback_text"].str.count(r"\?")
```

Step 12: Encode Sentiment

```
sentiment_map = {
    "Positive":2,
    "Neutral":1,
    "Negative":0
}

blinkit["Sentiment_Label"] = blinkit["sentiment"].map(sentiment_map)
```

Step 13: Encode Feedback Category

```
blinkit = pd.get_dummies(
```

```
    blinkit,
    columns=["feedback_category"],
    drop_first=True
)
```

Step 14: Encode Rating Category

```
blinkit = pd.get_dummies(
    blinkit,
    columns=["Rating_Category"],
    drop_first=True
)
```

Step 15: TF-IDF Features (Machine Learning)

```
from sklearn.feature_extraction.text import TfiblinkitVectorizer

tfiblinkit = TfiblinkitVectorizer(
    stop_words="english",
    max_features=100
)

text_features = tfiblinkit.fit_transform(blinkit["feedback_text"])

text_blinkit = pd.DataFrame(
    text_features.toarray(),
    columns=tfiblinkit.get_feature_names_out()
)
```

Merge

```
blinkit = pd.concat(
    [blinkit.reset_index(drop=True),
     text_blinkit.reset_index(drop=True)],
    axis=1
)
```

Step 16: Drop Unnecessary Columns

IDs usually do not help prediction.

```
blinkit.drop(
    columns=[
        "feedback_id",
```

```
        "order_id",
        "customer_id",
        "feedback_text",
        "feedback_date",
        "Month_Name",
        "Day_Name",
        "sentiment"
    ],
    inplace=True
)
```

Step 17: Check Final Dataset

```
blinkit.head()
blinkit.info()
```

Step 18: Save Engineered Dataset

```
blinkit.to_csv(
    "Blinkit_Feature_Engineered.csv",
    index=False
)
```

Features Created

Original Feature	New Feature
feedback_date	Year
feedback_date	Month
feedback_date	Day
feedback_date	Week
feedback_date	Quarter
feedback_date	Weekend
rating	Rating_Category
feedback_text	Feedback_Length
feedback_text	Word_Count
feedback_text	Character_Count
feedback_text	Sentence_Count
feedback_text	Capital_Letters
feedback_text	Exclamation
feedback_text	Question

Original Feature	New Feature
sentiment	Sentiment_Label
feedback_category	One-Hot Encoded Columns
feedback_text	TF-IDF Features